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Project Title:	Cardiac Atrophy and Diastolic Dysfunction During and After Long Duration Spaceflight: Functional Consequences for Orthostatic Intolerance, Exercise Capacity, and Risk of Cardiac Arrhythmias		
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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
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Zip Code:	75231-5129	**Congressional District:**	5
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Key Personnel Changes/Previous PI:	PI: Benjamin D. Levine, M.D. ; Co-PI: Michael W. Bungo, M.D.		
COI Name (Institution):	CoPI: Bungo, Michael (University of Texas, Houston)		
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Task Description:	Cardiac atrophy appears to develop during spaceflight or its ground based analogues, leading to diastolic dysfunction and orthostatic hypotension. Such atrophy also may be a potential mechanism for the cardiac arrhythmias recently identified in some crew members after long duration exposure to microgravity aboard the Mir space station. Recent work by the investigators of this flight experiment has suggested that cardiac atrophy may be progressive, without a clear plateau over at least 12 weeks of bed rest, and thus may be a significant limiting factor for extended duration space missions. The purpose of this experiment is to quantify the extent and time course of cardiac atrophy and identify its mechanisms. The functional consequences of this atrophy also will be determined for cardiac filling dynamics, orthostatic tolerance, exercise tolerance, and arrhythmia susceptibility both in space on the International Space Station, and following return to earth. Hypotheses: Hypothesis 1a: Cardiac atrophy occurs in humans after prolonged spaceflight as a function of both the basal hemodynamics of microgravity (reduced wall stress and volume load) and a reduction in physical activity. Hypothesis 1b: Cardiac atrophy develops progressively during long duration spaceflight, but eventually plateaus in response to a new steady-state of cardiac work. Hypothesis 2: Cardiac atrophy during prolonged spaceflight reduces ventricular distensibility manifested as decreased end-diastolic volume for any given filling pressure. We propose that this remodeling of the heart limits the utilization of diastolic suction and impairs ventricular relaxation and filling both at rest and during orthostatic stress. This impairment of both static and dynamic diastolic function, in combination with hypovolemia is the primary mechanism leading to a reduced stroke volume during orthostasis after spaceflight. Hypothesis 3: Prolonged spaceflight leads to alterations in ventricular conduction and refractoriness, and an increase in the frequency of arrhythmias. Objectives: Objective 1a: To determine the magnitude of left and right ventricular atrophy associated with long duration space flight, and to relate this atrophy to measures of physical activity and cardiac work in-flight. Magnetic resonance imaging (MRI) will be performed pre-and post-flight as the most accurate means of measuring cardiac mass, including sufficient repetitions to document recovery. Objective 1b: To determine the time course and pattern of progression of cardiac atrophy in-flight using cardiac ultrasound. Objective 2: To determine the functional importance of cardiac atrophy for orthostatic tolerance and the regulation of stroke volume, by using a novel combination of innovative, non-invasive imaging techniques to measure the dynamic component of diastole including: a) tissue Doppler imaging to quantify ventricular wall motion with high temporal resolution; b) color Doppler M-mode echocardiography to quantify intraventricular pressure gradients and assess the magnitude of diastolic suction; and c) magnetic resonance imaging with myocardial tagging to quantify the rate of untwisting. Objective 3: To identify changes in ventricular conduction and repolarization during and after long duration spaceflight, and relate these to changes in cardiac mass. Although the direct assessment of arrhythmia risk in otherwise healthy subjects is imprecise, we will obtain hypothesis generating, pilot data using the best non-invasive tests available by measuring: 1) delayed conduction via the signal averaged ECG; 2) dispersion of refractoriness from QT dispersion; and 3) microvolt T wave alternans during routine exercise. New Information Expected: After completion of this study, the clinical expression of cardiac atrophy during long duration space flight will be defined clearly, and its significance for orthostatic tolerance and cardiac performance, both systolic and diastolic, will be elucidated. In addition preliminary information will be obtained regarding ventricular conduction and repolarization that may provide either clinical reassurance, or pathophysiolgic insight into the risk for cardiac arrhythmias. Relevance to Space and/or Earth-Based Research: Once the magnitude, time course, and inciting factors for cardiac atrophy have been determined, effective countermeasures currently being developed by the investigators in parallel ground based experiments, may be applied, focused on normalizing cardiac work and volume during long-duration spaceflight. The investigators have recently demonstrated that preventing and/or reversing the atrophy and hypovolemia during short duration head-down tilt bed rest completely eliminates orthostatic intolerance after this ground based analogue. If a similar pathophysiology can be demonstrated after long duration space flight, a strong rationale can be made for implementing the same countermeasure during missions aboard the ISS, potentially eliminating a problem, namely post-flight orthostatic hypotension, that has plagued astronauts since the inception of the manned space program. The information obtained from these space flight experiments also may be relevant for patients after prolonged confinement to bed rest, or chronic reduction in physical activity, as well as for patients with disease processes that alter cardiac stiffness such as congestive heart failure, ischemic heart disease, and normal aging.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Once the magnitude, time course, and inciting factors for cardiac atrophy have been determined, effective countermeasures currently being developed by the investigators in parallel ground based experiments, may be applied, focused on normalizing cardiac work and volume during long-duration spaceflight. The investigators have recently demonstrated that preventing and/or reversing the atrophy and hypovolemia during short duration head-down tilt bed rest completely eliminates orthostatic intolerance after this ground based analogue. If a similar pathophysiology can be demonstrated after long duration space flight, a strong rationale can be made for implementing the same countermeasure during missions aboard the ISS, potentially eliminating a problem, namely post-flight orthostatic hypotension, that has plagued astronauts since the inception of the manned space program. The information obtained from these space flight experiments also may be relevant for patients after prolonged confinement to bed rest, or chronic reduction in physical activity, as well as for patients with disease processes that alter cardiac stiffness such as congestive heart failure, ischemic heart disease, and normal aging.

Task Progress:	The full experiment methods and scheduling are detailed in the Experiments Document for E377 also referenced as LS20447 (Revision B dated June 2009 is the most recent version). Briefly, data are collected pre-flight, in-flight, and post-flight according to the schedule presented therein for the purposes described above in the objectives. Cardiac magnetic resonance imaging is performed on a 3T magnet and data is acquired for tissue imaging (cardiac mass), but also with tagging (to assess myocardial dynamics of untwisting), spectroscopy (to assess triglyceride/fat content), and delayed gadolinium enhancement (to assess fibrosis). Cardiac ultrasound (echocardiography) is performed to correlate with MRI measurements but also to assess for cardiac function (particularly as related to a number of Doppler parameters of diastolic function) and to track any temporal changes over the duration of the data collection period. A separate session of echocardiography is performed to assess the cardiovascular response to increased work loads and impact upon mission duties. Holter monitoring is performed with a high frequency response device. This allows extrapolation of the data beyond arrhythmia detection to include sophisticated analysis of signal averaged ECG, microvolt T wave alternans, QT dispersion, and emerging techniques of T wave analysis (repolarization) to reflect the risk of possible malignant dysrhythmia. Ambulatory blood pressure monitoring and Acti-watch monitoring are used to estimate cardiac work and physical work respectively and to aid in the assessment of orthostatic tolerance in the post-flight period. Tilt table testing with attendant hemodynamic and ultrasound monitoring is used to characterize responses to simulated Moon, Mars, and Earth gravity. To date, three crewmembers have completed the full protocol, a fourth is nearing completion, and a fifth is in active data collection. Several in-flight data sessions fell outside the “windows” specified in the experiment document. Some post flight data was also outside requested time constraints. One pre-flight MRI scan was unable to be obtained with the gadolinium delayed enhancement. Final impact of these anomalies will not be sufficiently defined until all data is collected and analyzed, since the impact of biologic variability versus temporal variability will not be defined until those data are available; however, at this time, it is not expected that these anomalies will have critical impact upon the ability to derive sufficient conclusions from data analyses. On November 2, 2009, a significant hardware failure occurred that resulted in the ISS Ultrasound equipment (Echocardiograph) becoming inoperable. However to the great credit of the entire JSC team, the problem was identified, new circuit boards quickly manifest, flown on the next available shuttle (STS-129), and the problem fixed by a dedicated, committed crew. This remarkable team effort allowed us to obtain virtually all of our required data without significant loss. As the limited discussion presented above details, each technique contributes to several objects and each objective can be at least partially achieved by several of the techniques. This design allows for significant data “usability”. Thus, deviations from the initially proposed schedule and the initially proposed techniques will have impact upon the fidelity of conclusions that may result from the project, but the overall impact may be mitigated by the successful completion of other components of the study and protection against “catastrophic” failure ensured. Data analysis is ongoing. We are reluctant to analyze too much data prior to completion of all studies by an individual astronaut so that we can sustain blinded analysis and avoid bias. Nevertheless, each data set (echo imaging, MR imaging, rhythm and BP monitoring) have been analyzed in part to determine data quality. Once complete data sets are obtained on individual subjects, complete analysis will be able to be finished expeditiously. With deference to subject confidentiality issues, preliminary findings have been presented to NASA management but are withheld in this report.
Bibliography Type:	Description: (Last Updated: 03/19/2026)
Articles in Peer-reviewed Journals	Shibata S, Perhonen M, Levine BD. "Supine cycling plus volume loading prevent cardiovascular deconditioning during bed rest." Journal of Applied Physiology. 2010 May;108(5):1177-86. Epub 2010 Mar 11. PMID: 20223994 , May-2010
Articles in Peer-reviewed Journals	Fu Q, VanGundy TB, Galbreath M, Shibata S, Jain M, Hastings JL, Bhella PS, Levine BD. "Cardiac origins of the postural orthostatic tachycardia syndrome." Journal of the American College of Cardiology, 2010 Jun 22;55(25):2858-68. PMID: 20579544 , Jun-2010